

All India Coordinated Research Project on Agrometeorology
अखिल भारतीय समन्वित कृषि-मौसम विज्ञान अनुसंधान परियोजना
(AICRPAM)

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दिसंबर 2024 के दौरान जारी मौसम की स्थिति और कृषि मौसम संबंधी सलाह

Weather conditions and Agromet advisories issued during December 2024

1. सामान्य मौसम की स्थिति

1. General Weather Conditions

01 अक्टूबर से 31 दिसंबर 2024 के दौरान, पूरे देश में 117 मिमी बारिश हुई है, जो इसी अवधि के लिए सामान्य वर्षा (121 मिमी) की तुलना में -3% कम है। 36 मौसम विज्ञान उप-विभागों में से, 02, 06, 14, 10 और 04 उप-विभागों में क्रमशः बहुत अधिक, अधिक, सामान्य, कम बारिश और बहुत कम बारिश हुई, जैसा कि चित्र 1 में दर्शाया गया है।

During 01 October - 31 December 2024, the country as a whole has received 117.0 mm of rainfall, which is -3% deficit, compared to the normal rainfall (121 mm) for the same period. Out of 36 meteorological sub-divisions, rainfall was large excess, excess normal and deficient in 02, 06, 14, 10 and 04 sub-division (s), respectively as depicted in Fig. 1.

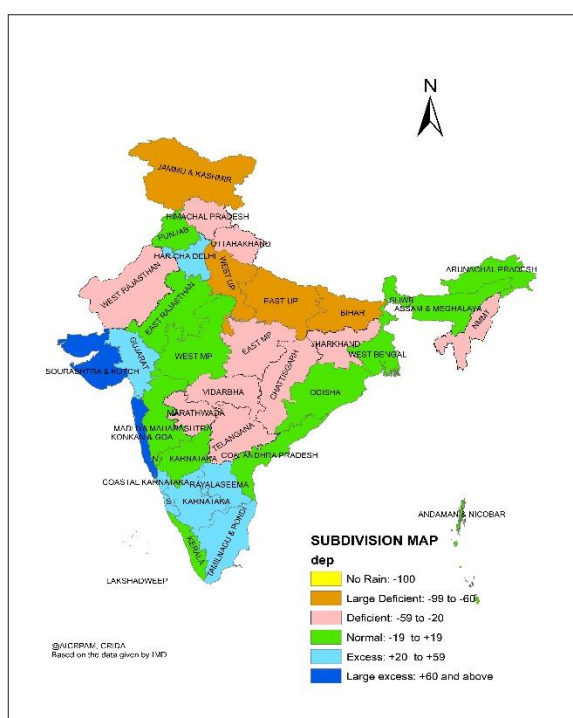


Fig. 1: Seasonal rainfall status (01 October – 31 December 2024) in 36 meteorological sub-divisions, India

1. Agromet advisories issued by AICRPAM centres during December 2024

Date of issue	Advisory issued for the region	Agromet Advisory issued
06-12-2024	Punjab	- Irrigate all the vegetables in the field as the weather is expected to be dry during second week of December. It is the optimum time for sowing winter vegetables like carrot, radish, turnip, spinach etc. - Apply irrigation to ber orchards as the trees are loaded with fruits. Irrigation should be done at 3–4-week intervals. - To protect the young fruit plants from upcoming winters, arrangements should be made to prepare the thatches or kullies of sarkanda or farm waste materials on the plants during November.
	Assam	- Due to favorable weather appearance of the Sigatoka leaf spot may be observed in the Banana plant. Spray Carbendazim @ 0.5 gm/litre of water at first appearance followed by 4 numbers of sprays at 30-day intervals. Maintain the banana orchard weed-free and avoid water stagnation. - Harvesting and threshing of already matured Sali paddy may be continued in sunny weather. Harvesting at the right time is crucial to prevent losses and ensure good grain quality and high market value. - Sawfly infestation can be seen in many vegetative toria fields; to control, spray Chlorantraniliprole 18.5 SC @ 0.2 ml or Neemacin 1500 ppm @ 3 ml per litre of water in early morning.
	West Bengal	- Raised bed cultivation/ridge & furrow planting helps onion plants to survive future excess rainfall and moisture conditions. In case of excess moisture in the field, farmers may prepare the seedlings in the seed bed and then transplant the seedlings @10-12 kg/ha. Pusa red, N-53, Bhima Super, Agrifound light red are the popular options within onion varieties. - Sowing of mustard can be done after the kharif rice harvesting. The recommended seed rate is given 4-5 kg/ha. Pusa Mustard 25 (NPJ 112), RH 749, NRCDR-2, and PM-28 are some of the promising mustard varieties that farmers may adopt. - The cattle sheds should be covered from all sides to prevent cold air intrusion in the shed. The shed should be cleaned periodically to avoid any bacterial and fungal infestation

	Jharkhand	• Timely sown chickpea is in the germinating stage. Farmers are advised to go for first irrigation if available moisture is not enough in the soil for germination. • When transplanted cabbage and cauliflower seedlings are established in three weeks, go for hoeing, weeding and apply 55 kg urea per acre followed by irrigation.
	Himachal Pradesh	• Use improved mustard varieties like Pusa Jai Kisan, Pusa Bold, Vardan, Varuna, Kranti Chhattisgarh Mustard, etc., for sowing at a seed rate of 4-5 kg per hectare. • Looking into the decrease in temperature and its possible adverse effects, dairy farmers are advised to keep the animals in the cattle shed with a concrete/brick roof. • Raise the seedlings of winter vegetables like cauliflower, cabbage, chilli, tomato, brinjal, etc. • To control Alternaria blight in pigeon pea crops, it is advised to spray Mancozeb 75 WP @ 2 g or Carbendazim 50 WP @ 1g per litre of water.
	Chhattisgarh	• Farmers with irrigation facilities should irrigate their rice fields as required. Irrigation should be withheld in the fields which are in saturation. • Farmers are advised to control Armyworm and Panicle mite spray fungicides when the weather is clear. To control Sheath Blight, farmers are advised to spray insecticides when the weather is clear. • The field where the rice crop is approaching maturity and the field is having some moisture; farmers can go for relay cropping (utera) of linseed or lathyrus.
13-12-2024	Kerala	• To control leaf spot in bananas, rhizome treatment in 50 g Pseudomonas per litre water for 30 minutes; spray Pseudomonas 20 g per litre at initial stages, or a 1% Bordeaux mixture can be sprayed 3 to 4 times in a 3 to 4-week interval. • Apply 1% Bordeaux mixture or 0.3% copper oxychloride solution in the heart of the crown. Apply lime. Spray borax also on leaves @ 5g per litre. Also adopt all the basal management practices as described for the yellowing, above.

13-12-2024	Tamil Nadu	• In Paddy (crop establishment), Maintain the water level to avoid crop damage. • In sorghum and maize, Continuous rainfall and wind may affect the crop stand; hence farmers are requested to practice earth up and form new ridges to provide additional anchorage to the plants. • Due to water stagnation, yellowing of leaves is noticed. Hence, foliar spray of 2% DAP + 1% KCl (MOP) or TNAU pulse wonder @ 2 kg/acre may be given. • Since rainfall with wind speed >10 kmph is expected, provide propping to early season sugarcane. Create drainage facilities. • High wind speeds may damage the physical appearance of the crop. Hence, farmers are advised to practice propping. Earthing up should be practiced to avoid lodging in banana. • In cotton, Crop residue mulching may be done to reduce soil erosion. Precaution may be taken to avoid water stagnation in the field. Immediate action for earth up and forms new ridges to prevent lodging of crops due to continuous rainfall.
	Bihar	• Farmers are advised to complete the sowing of wheat. Improved varieties: - <i>Raj-3077, WH-147, G.W.-190, G.W.-322, G.W.- 273, H.I.- 1544, Raj- 4037, H.I- 8498, Lok-1 and Raj-3765</i>. Seed rate: - 100 kg seed per ha. Application of Chlorpyriphos (20 EC) @ 5 litres per ha. with pre-irrigation is recommended before sowing in the fields having perennial problems of termite infestation. The recommended dose of fertilizer for N: P: K is 120, 50 and 40 kg/ha. • Complete the sowing of barley in this week. Improved varieties: <i>R.D-2035, R.D-2786, R.D-2715, R.D-2849 and B.H.- 902</i>. Application of Chlorpyriphos (20EC) @ 5 litres per ha. with pre-irrigation is recommended before sowing in the fields having perennial problems of termite infestation. • To control blight in gram. Spray of Mencozeb @ 2g/ litre of water.

20-12-2024	Marathwada	• Due to cloudy weather conditions, if an infestation of pod borer & pod fly is observed in pigeon pea crop. Make use of T-shaped bird perches @ 20 per acre and 3 pheromone traps per acre. For management of pod borer in pigeon pea take a spray of Emamectin Benzoate 5% @ 4.4 g or Spinosad 45% @ 3 ml or indoxacarb 14.5% @ 8 ml or Chlorantraniliprole 18.5% @ 3 ml per 10 litres of water. For management of pod fly take a spray of Lambda cyhalothrin 5% @ 8 ml or Lufenuron 5.4% @ 12 ml per 10 litres of water. • For management of Rhizome fly in turmeric take a spray of Quinalphos 25 % @ 20 ml or Dimethoate 30 % @ 15 ml per 10 litres of water with an interval of 15 days during clear weather conditions. Cover the rhizome with the help of soil. (As the label claim was not given by the Central Insecticides Board for Rhizome fly in turmeric the research recommendation of a university is mentioned here). For management of leaf spot disease in the Turmeric crop take a spray of Azoxystrobin 18.2% + Difenoconazole 11.4% (Premixed Fungicide) @ 10 ml + 5 ml Sticker per 10 litre of water during clear weather conditions. As per requirement irrigation management should be done in turmeric crop.
	Madhya Maharashtra	• Foliar spray of 2% Urea for chickpea under moisture stress conditions is advised and a second spray of 2% Potassium Nitrate after 10-15 days after the first spray is advised. • Foliar application of 2% Potassium Nitrate on Sorghum for reduction of severity of moisture stress is advised. • For control of Alternaria leaf spot disease of safflower, spraying of combi-fungicide Carbendazim (12%) + Mancozeb (63%) @ 2 gm per liter of water is suggested. • For control of pod borer of chickpea, spraying of biopesticide HANPV (<i>Heliokil</i>) @ 1 ml per litre of water or Insecticide Chlorantraniliprole 18.5 % S.C. 2.5 ml per 10 litre of water is recommended

	Konkan	• The tree's roots need to be stressed for the mango to bloom. For this, one meter of grass around the trunk of the mango orchards should be removed to pulverize the soil in the basin, shallow ploughing should be done in the open space in the mango orchard, and the orchard should be cleaned as soon as possible so that the amount of water in the soil can be reduced quickly and help the tree bear stress. • Avoid consecutive spraying of the same insecticide. Clean the cashew orchard by removing weeds and then use the pesticide spray. Additionally, spray nearby native trees along with the grafted cashew trees to achieve effective control over the pest infestation. • Due to cloudy weather, it likely led to an outbreak of anthracnose disease. The disease causes irregularly shaped hazel spots on the leaves and if the disease intensifies, the whole leaves blight off and fall off. To control disease, spray Mancozeb 20 gm per 10 litres of water.
	Gujarat	• Prop the sugarcane by tying together the canes in clumps to protect against high wind speed. Remove 25% lower mature leaf to improve sugar content and yield in sugarcane at 6 months of sowing. • Undertaking the nipping by plucking the apical buds of the crop at about 30 to 40 DAS is done to stop the apical growth and improve the yield of the gram. • Install pheromone traps @ 5/ha for monitoring of pod borer of pigeon pea. To control it, spray Emamectin Benzoate 5% @ 4.4 g or indoxacarb 14.5% @ 8 ml or Spinosad 45% @ 3 ml per 10 litres of water. Sterility mosaic disease is transmitted by mites, so to control it spray Propargite 57 % EC 10 ml 10 litre of water.